Aprendizaje profundo para el análisis de imágenes biomédicas

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Seminario – Cátedra de Inteligencia Computacional 2020

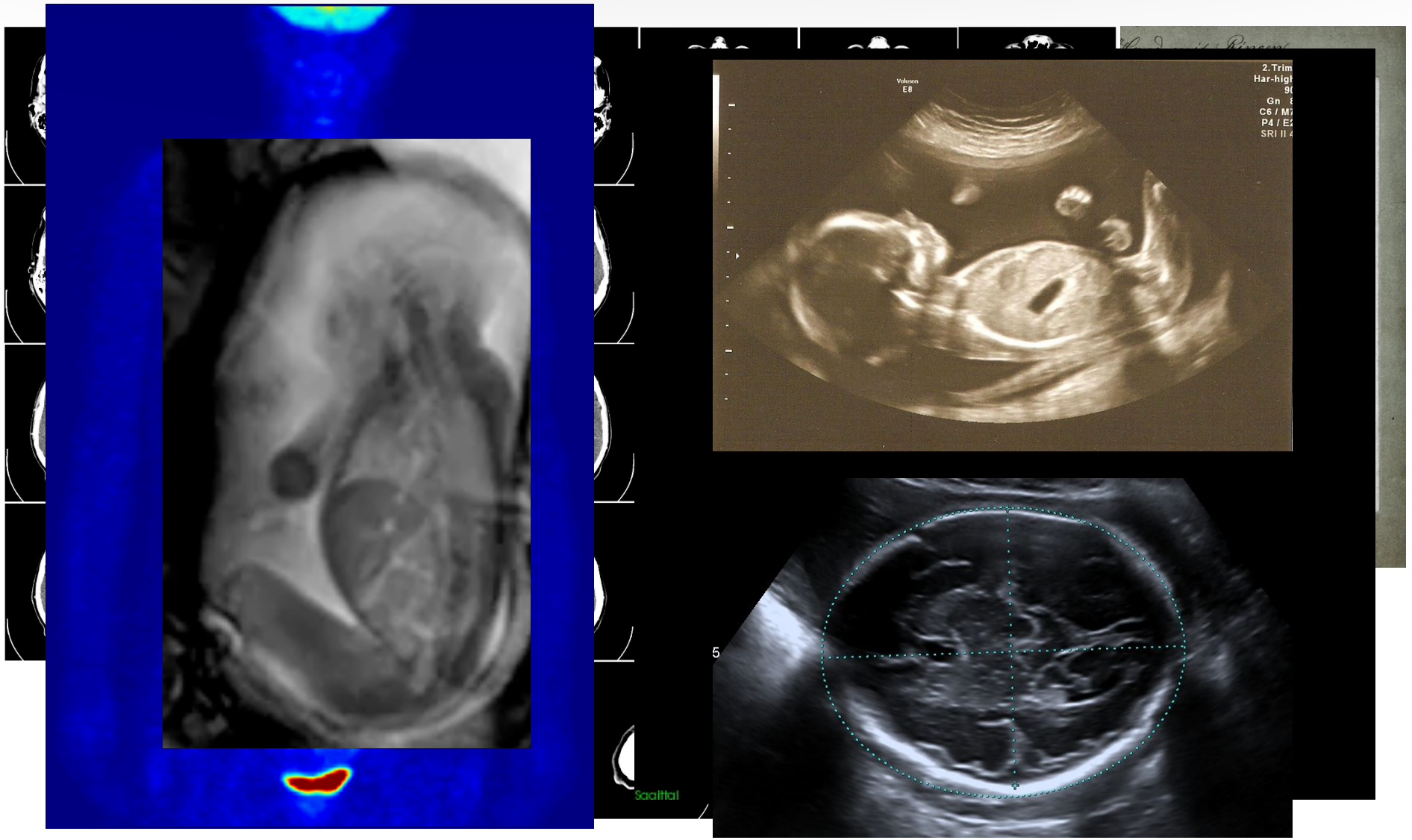
Imágenes Médicas

Imágenes Médicas

"imágenes utilizadas para crear representaciones del cuerpo humano, o partes de él, con propósitos clínicos o para la investigación en ciencia médica …"

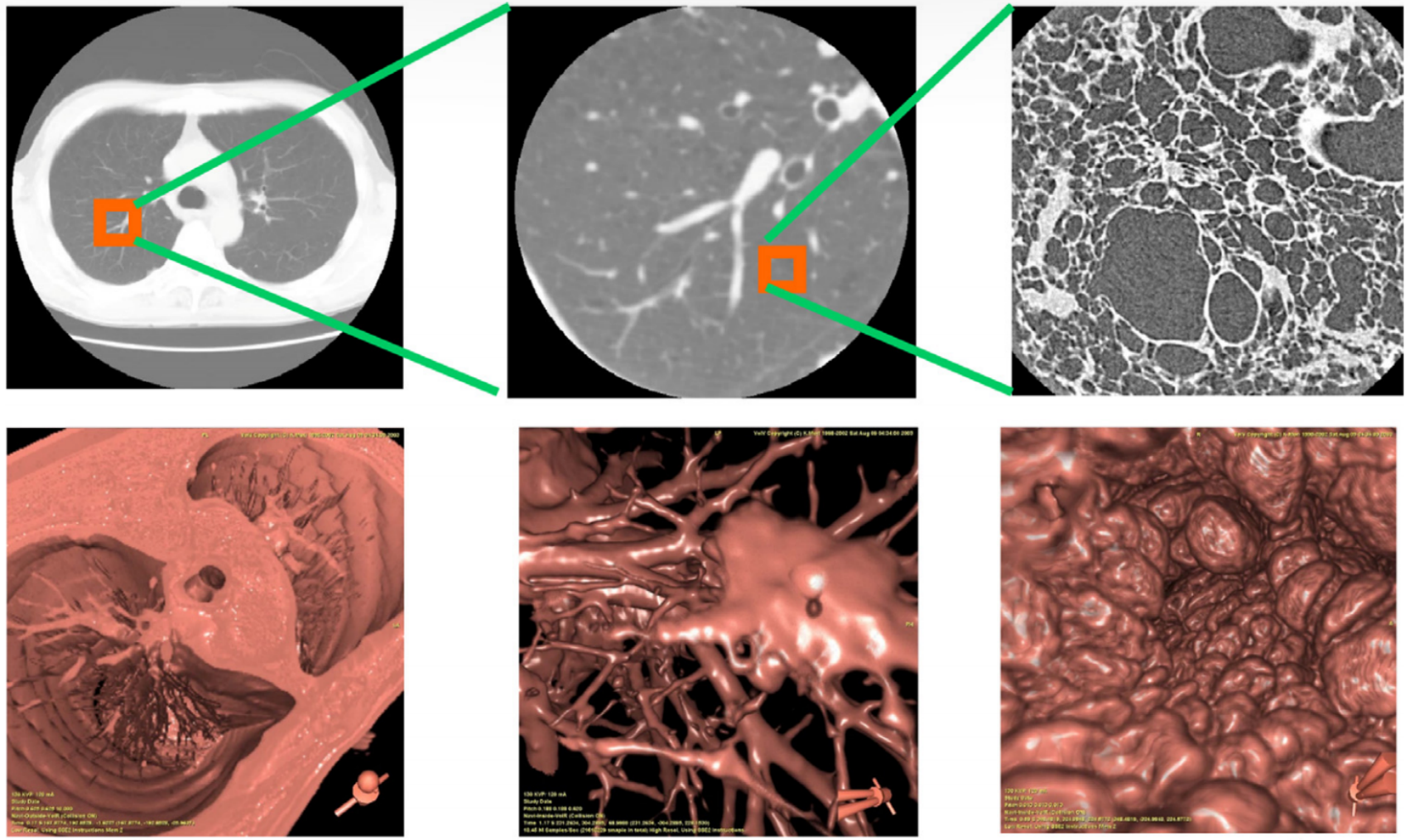
Imágenes Médicas

Múltiples modalidades



Imágenes Médicas

Múltiples escalas



Imágenes Médicas

Múltiples dimensiones

2D

Imágenes Médicas

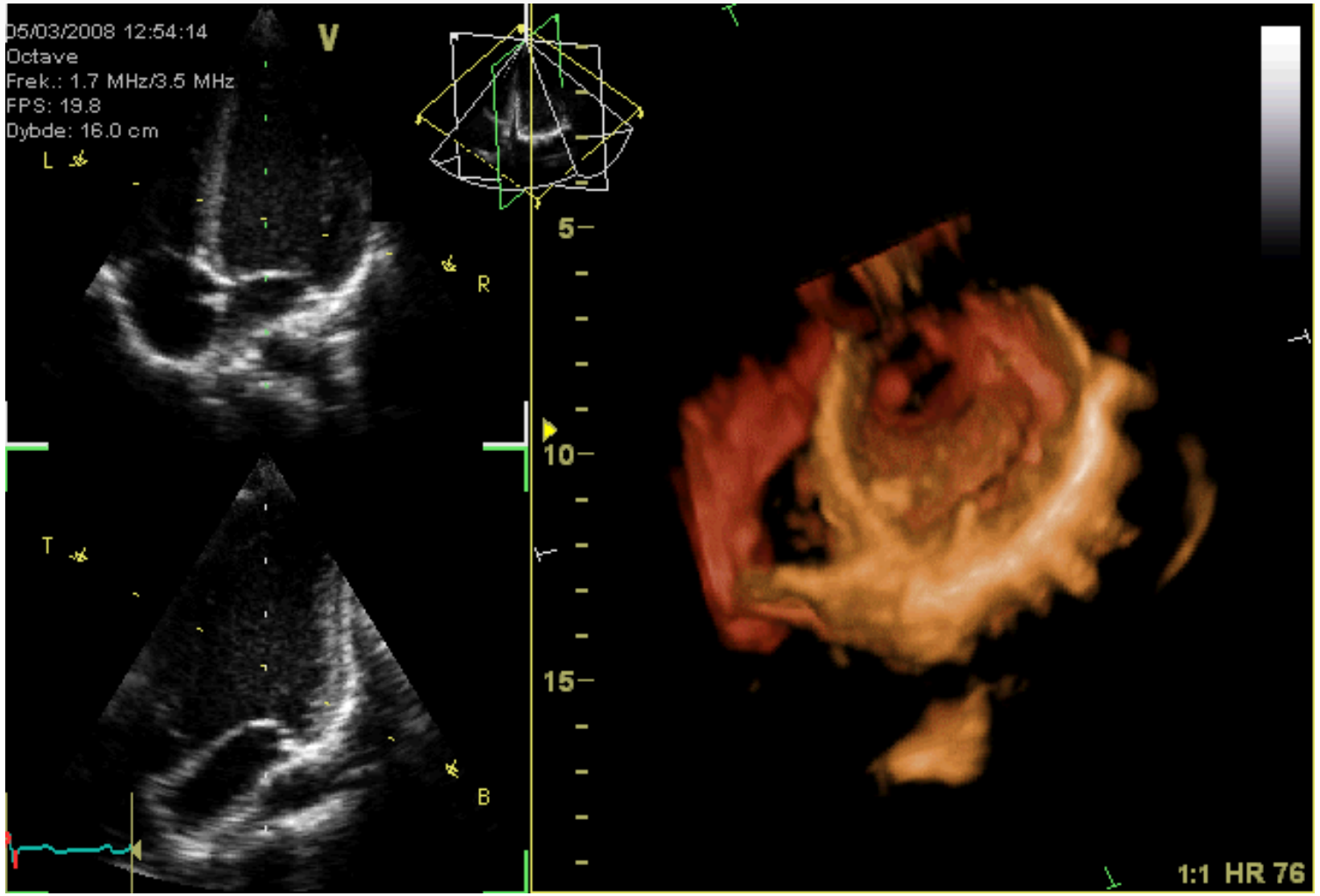
Múltiples dimensiones

3D

Imágenes Médicas

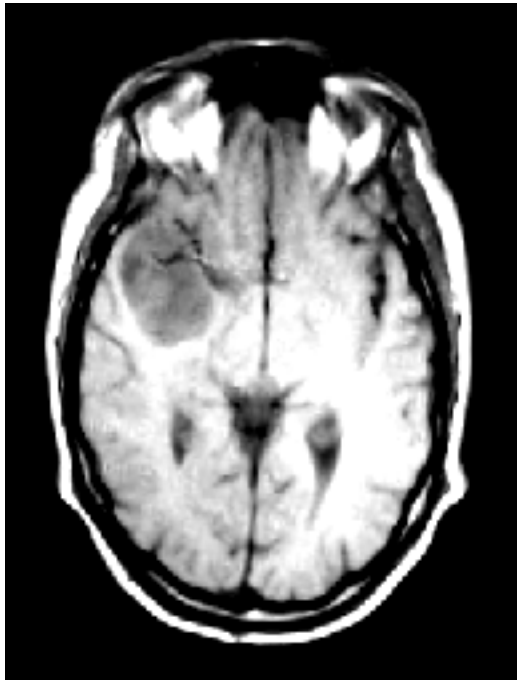
Múltiples dimensiones

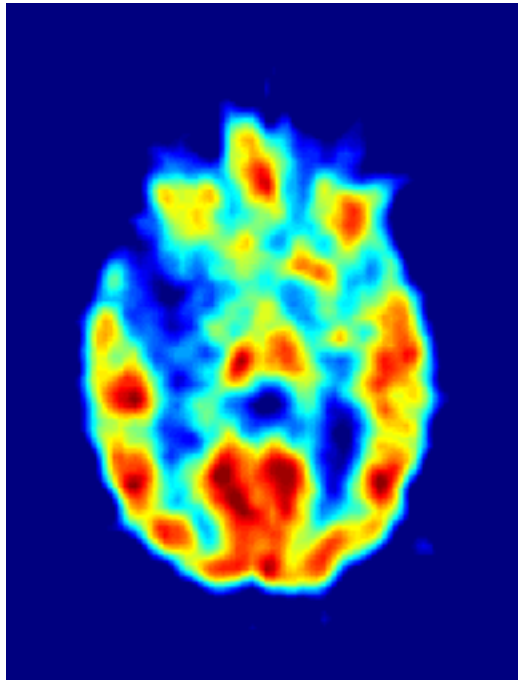
4D

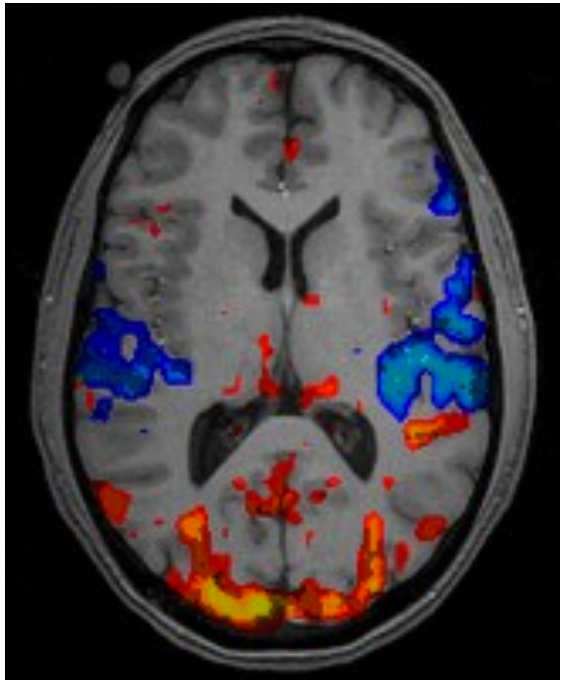


Imágenes Médicas

Anatómicas y funcionales







Análisis de Imágenes Médicas

Análisis de Imágenes Médicas

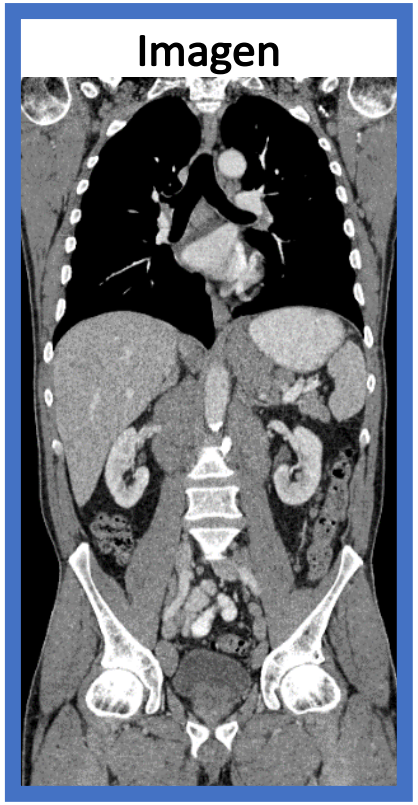
- Área altamente **interdisciplinar** (medicina, física, ciencias de la computación, biología, matemática)

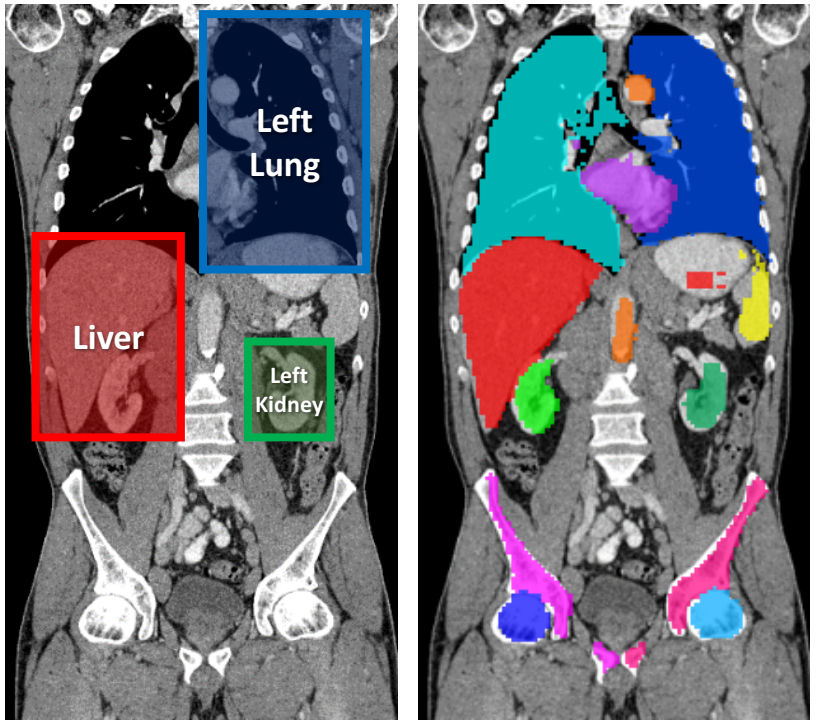
- Focalizada en el desarrollo de **métodos y técnicas** (mayormente computacionales) para la **extracción de información útil** a partir de dichas imágenes

- El objetivo último es contribuir con herramientas que faciliten:
  - la **investigación** biomédica,
  - el **diagnóstico** de enfermedades,
  - el **seguimiento** de terapias y
  - las **intervenciones** quirúrgicas

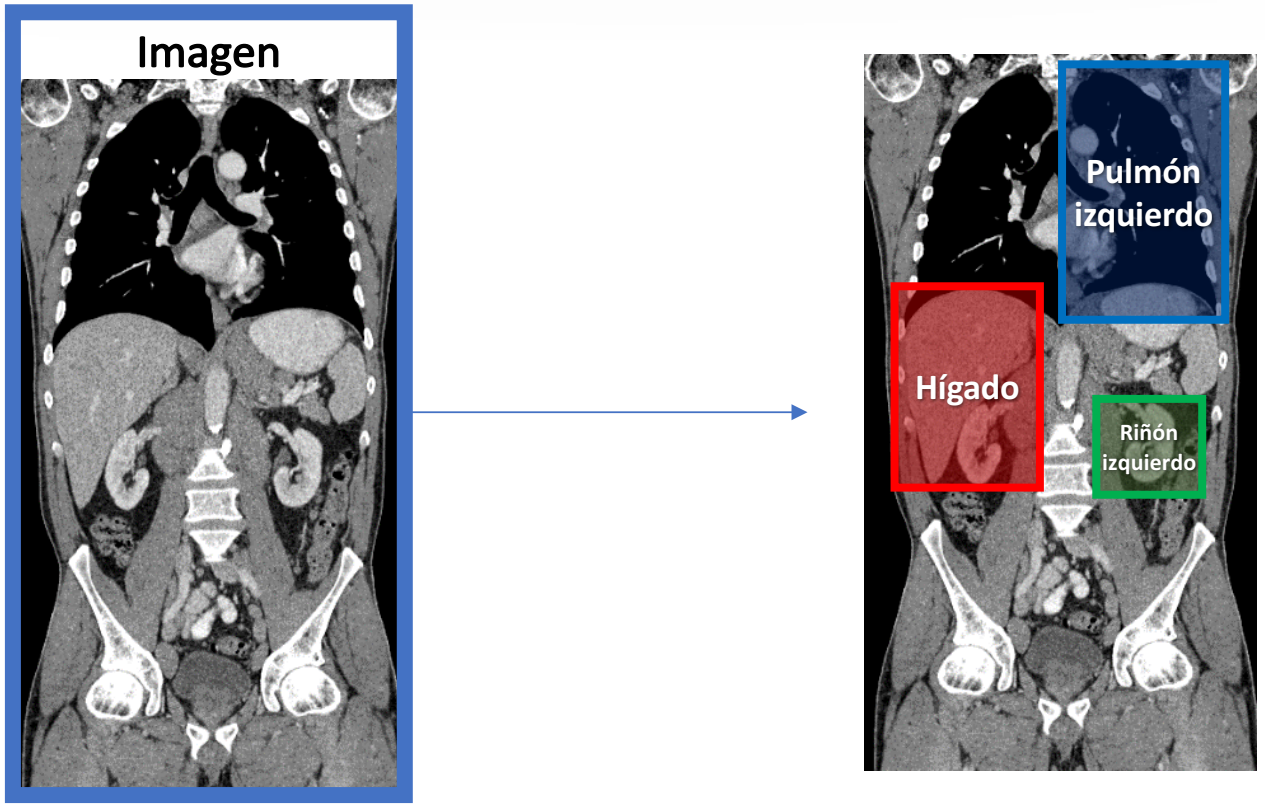
Métodos Computacionales para el Análisis de Imágenes Médicas

Problemas clásicos de visión computacional

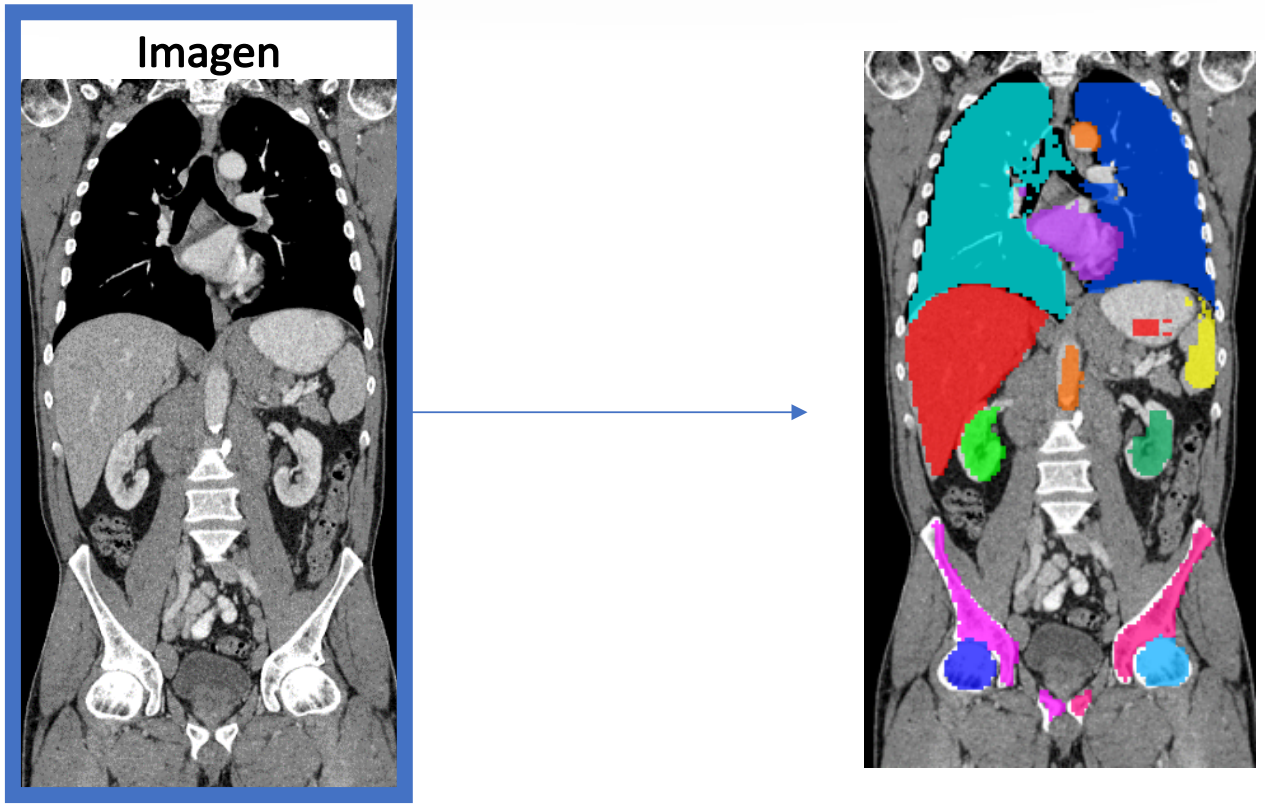




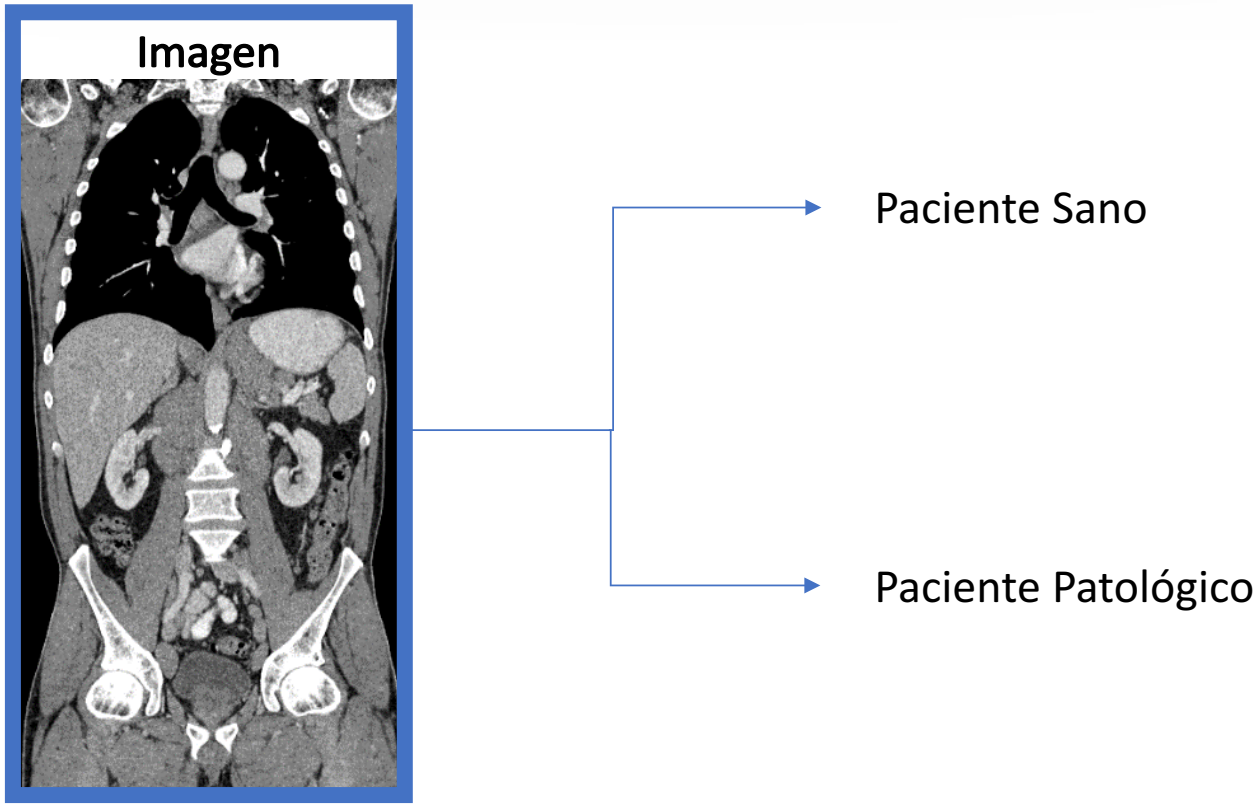
Detección de objetos



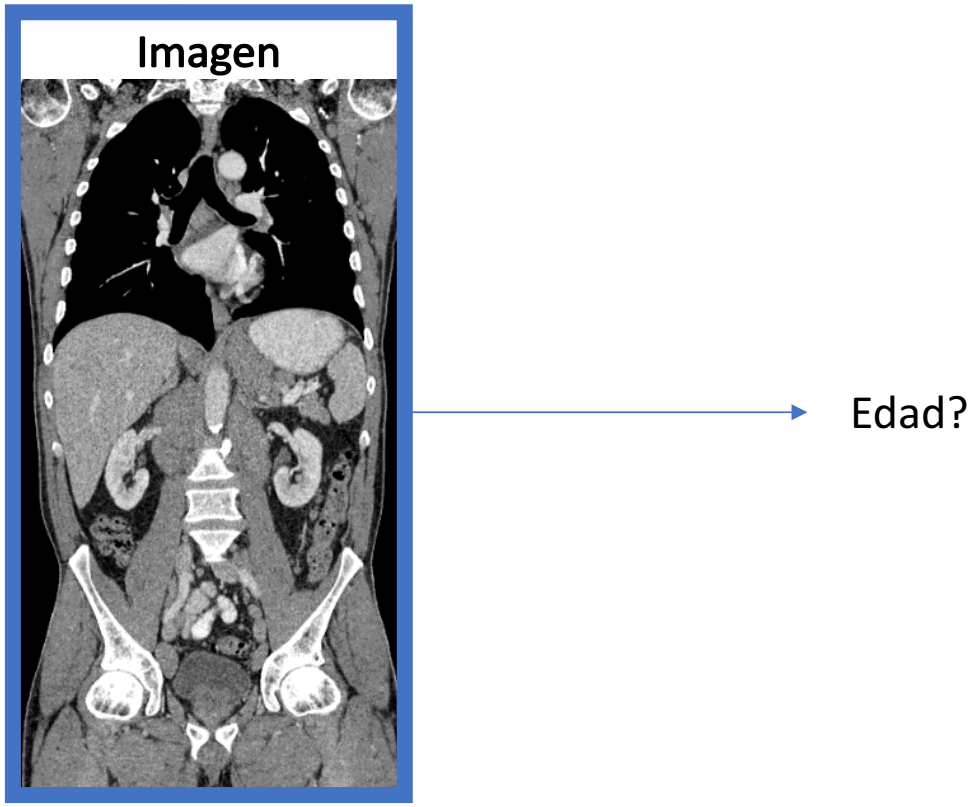
Segmentación



Clasificación



Regresión



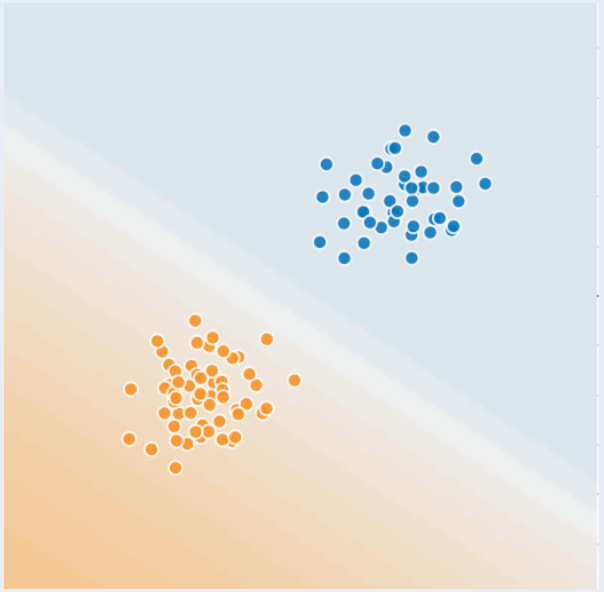
Aprendizaje Profundo para el Análisis de Imágenes Médicas

Enfoque tradicional

en aprendizaje maquinal supervisado

”Artesanal”

$(c_0, c_1, c_2 \ldots. c_n)$



¿Bicicleta?

Extracción de características

Selección de características

Clasificador

Predicción

Aprendizaje profundo

Una técnica para implementar aprendizaje maquinal

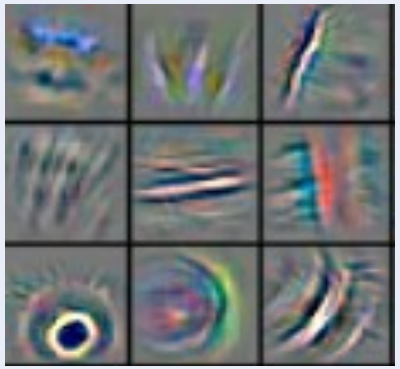
Los modelos basados en deep learning son capaces de aprender **representaciones** de los datos de entrenamiento en **múltiples niveles de abstracción** (capas), componiendo módulos simples que sucesivamente transforman dichas representaciones en otras con mayor nivel de abstracción.

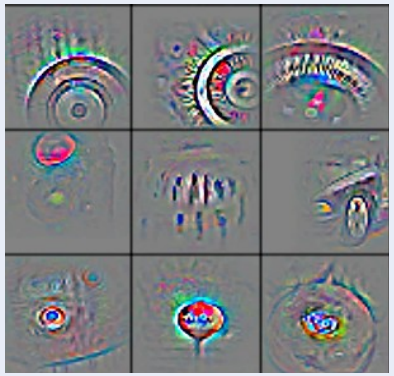
Aprendizaje profundo

Multiples niveles de abstracción

Único modelo entrenado ”end to end”

Bordes → Formas → Partes → Clasificador → ¿Bicicleta?

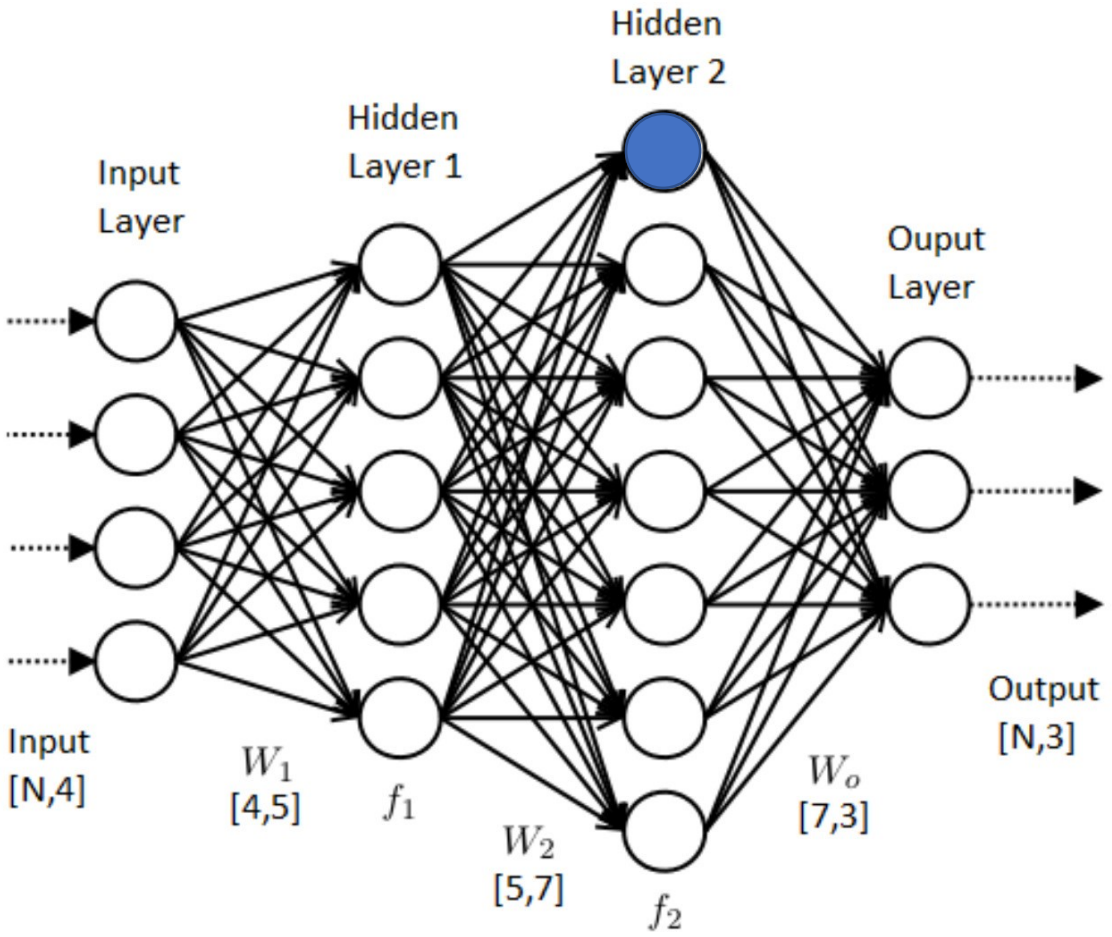




?

?

Redes Neuronales Artificiales



Cómo procesamos una imagen con un perceptrón?

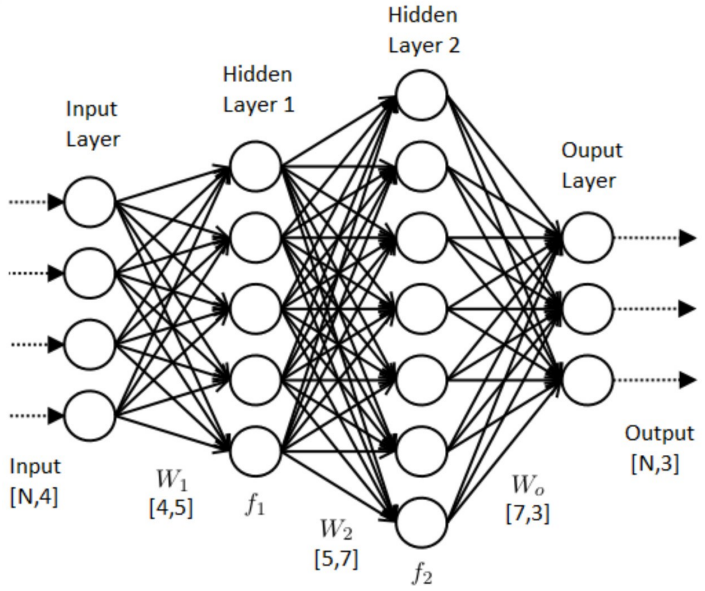
90	91	92	93	94	95	96	97	98	99
80	81	82	83	84	85	86	87	88	89
70	71	72	73	74	75	76	77	78	79
60	61	62	63	64	65	66	67	68	69
50	51	52	53	54	55	56	57	58	59
40	41	42	43	44	45	46	47	48	49
30	31	32	33	34	35	36	37	38	39
20	21	22	23	24	25	26	27	28	29
10	11	12	13	14	15	16	17	18	19
0	1	2	3	4	5	6	7	8	9

1024 x 1024

Imagen vectorizada
o extracción manual de características

N^2 (img vect) o M (features)

~ 1 Million de componentes



Perceptrón multicapa

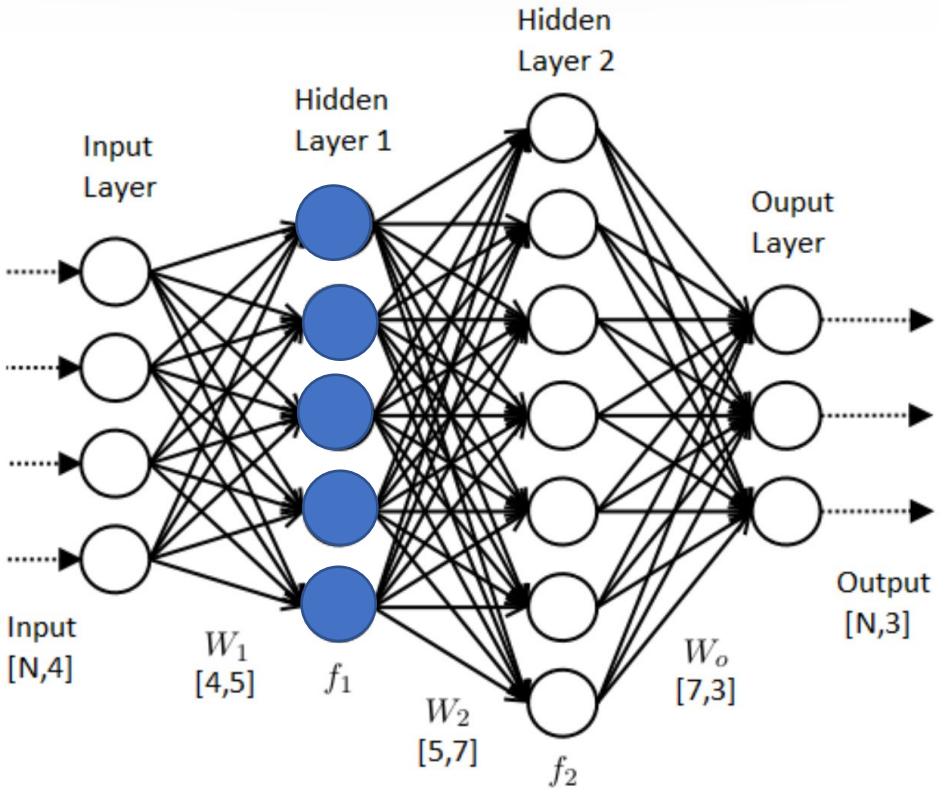
Cómo procesamos una imagen con un perceptrón?

~ 1 Millón de conexiones

x

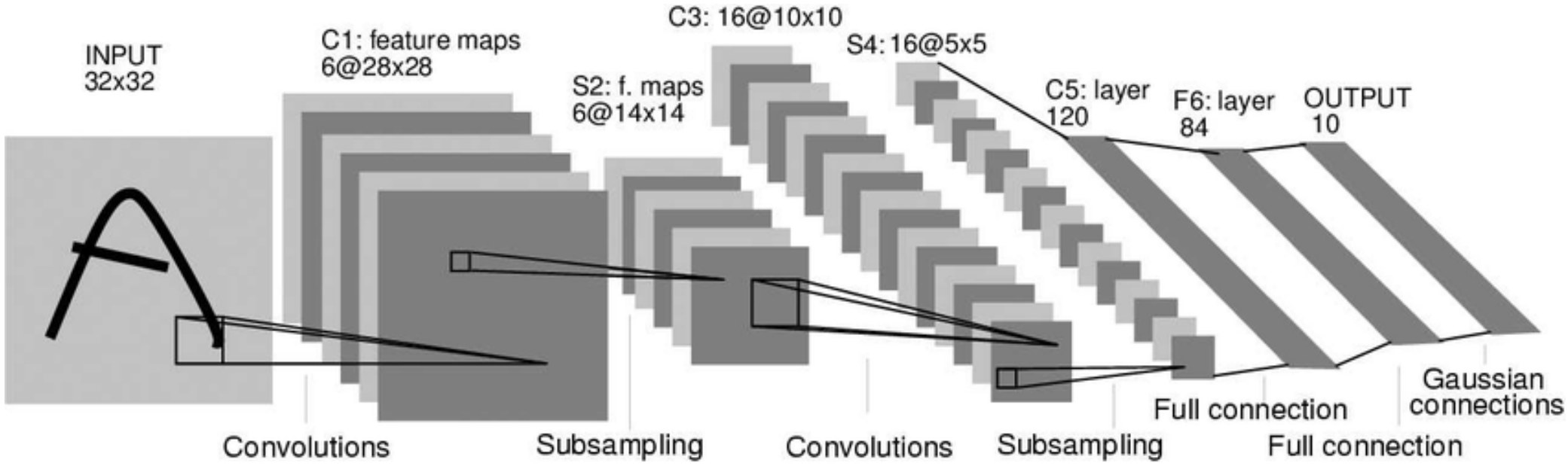
1024 neuronas ocultas

~ 1024 Millones de parámetros



Redes neuronales convolucionales

(Lecun 1998)



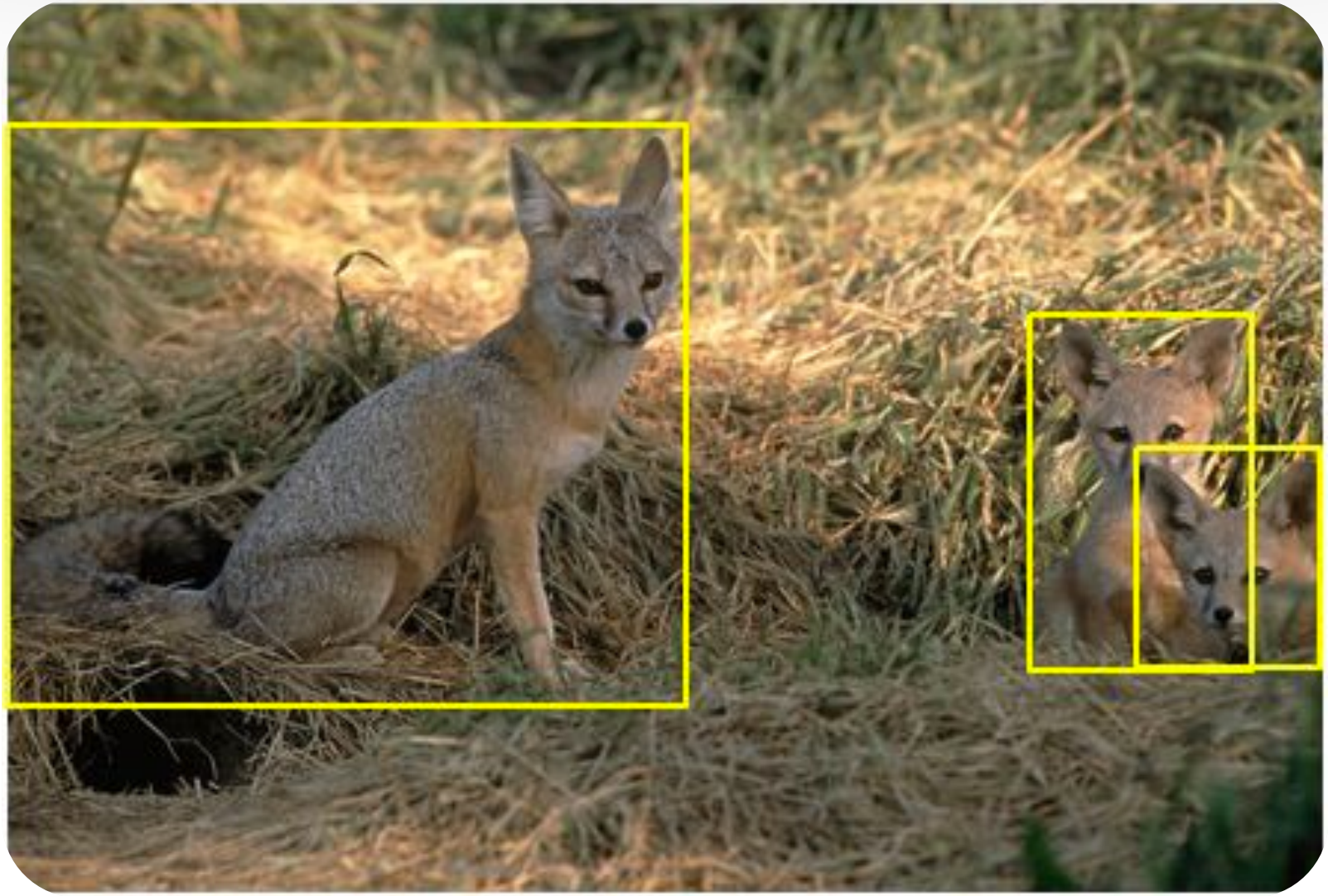
Redes neuronales convolucionales

ImageNet Challenge 2012
Clasificación de imágenes y detección de objetos

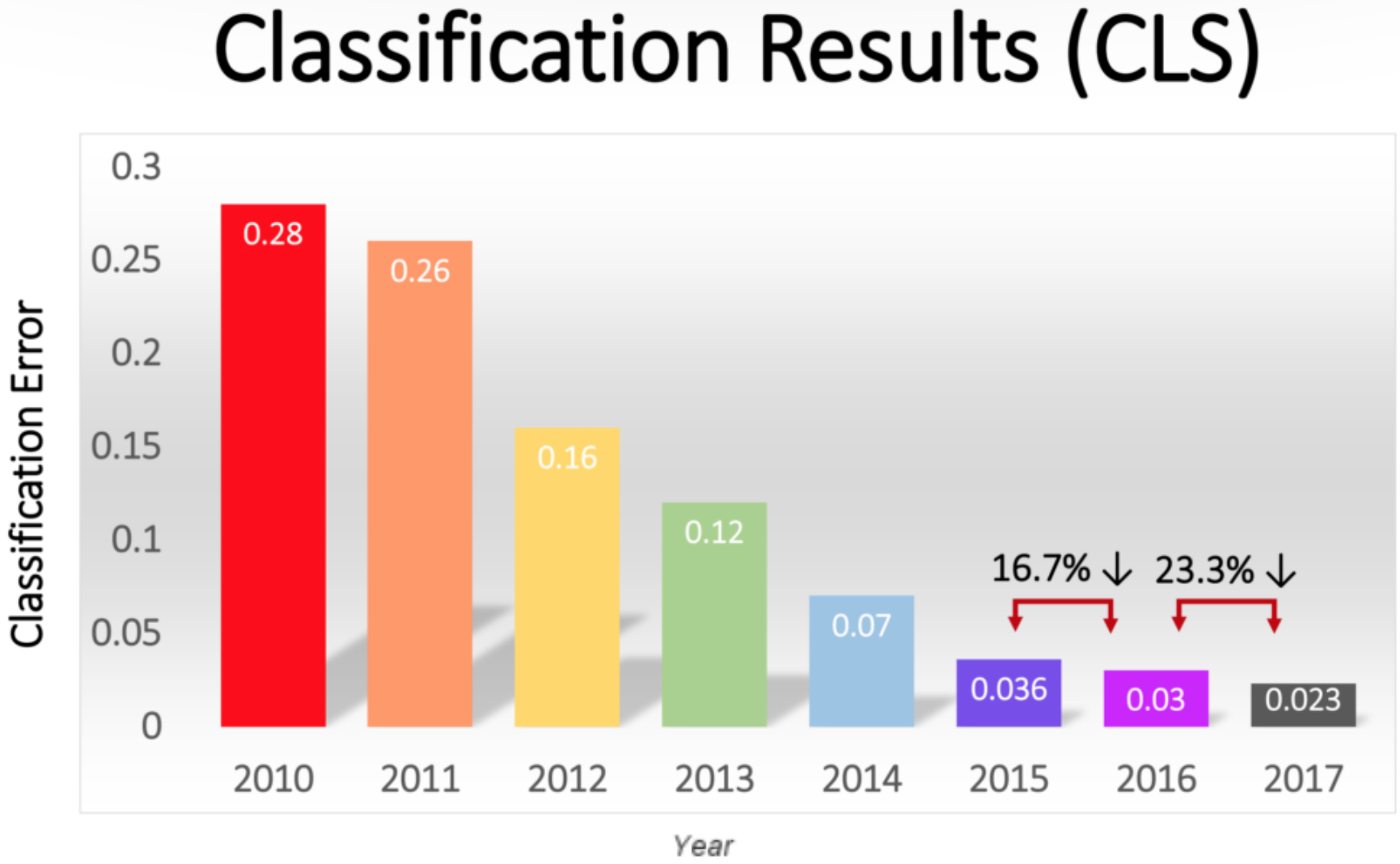
ImageNet Challenge

15.000.000 de imágenes anotadas para ser clasificadas en 1000 categorías

ImageNet Challenge



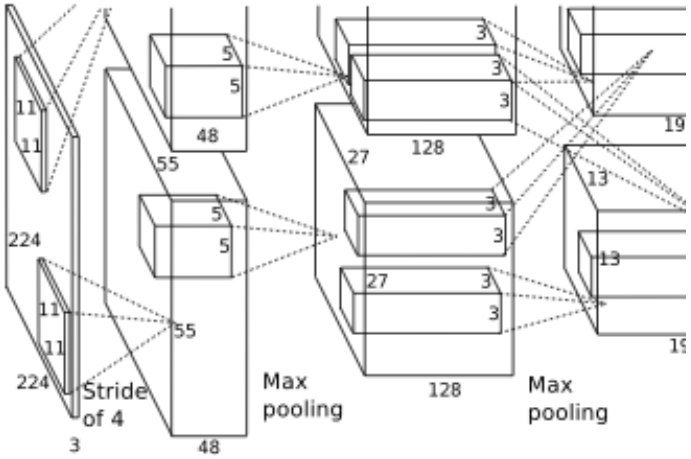
ImageNet Challenge



Computer Vision Revolution

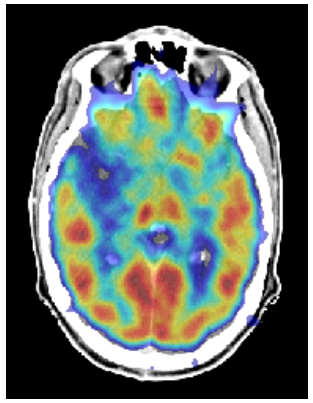
Más datos

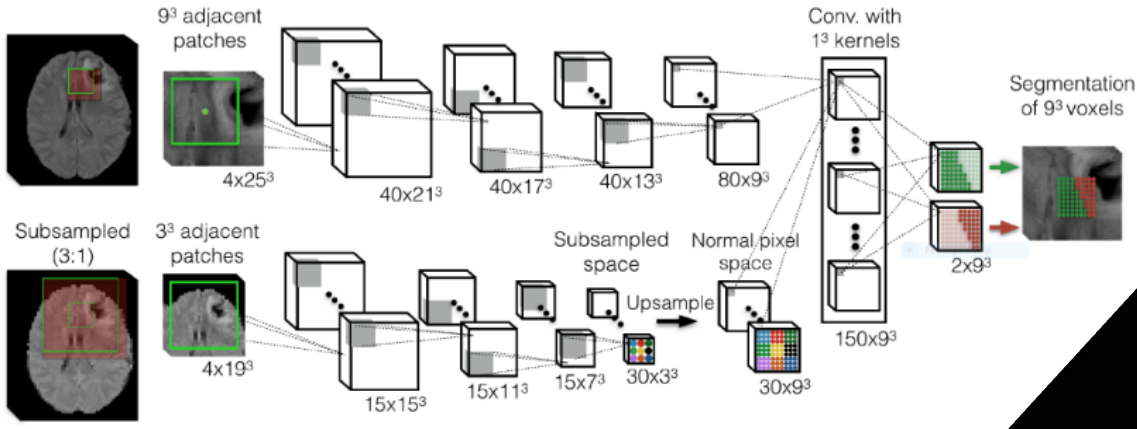
Mejores modelos

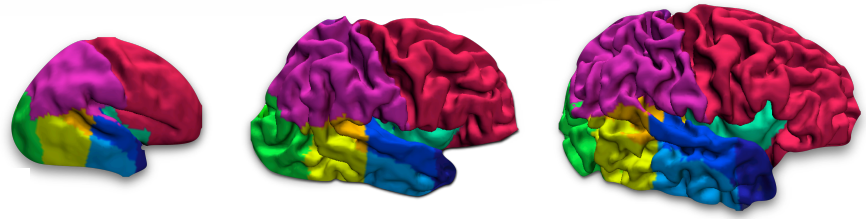


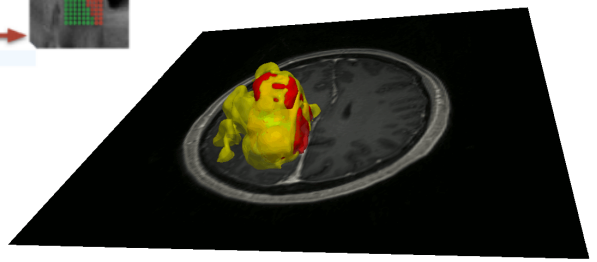
GPUs más poderosas

Deep ConvNets para Análisis de Imágenes Médicas

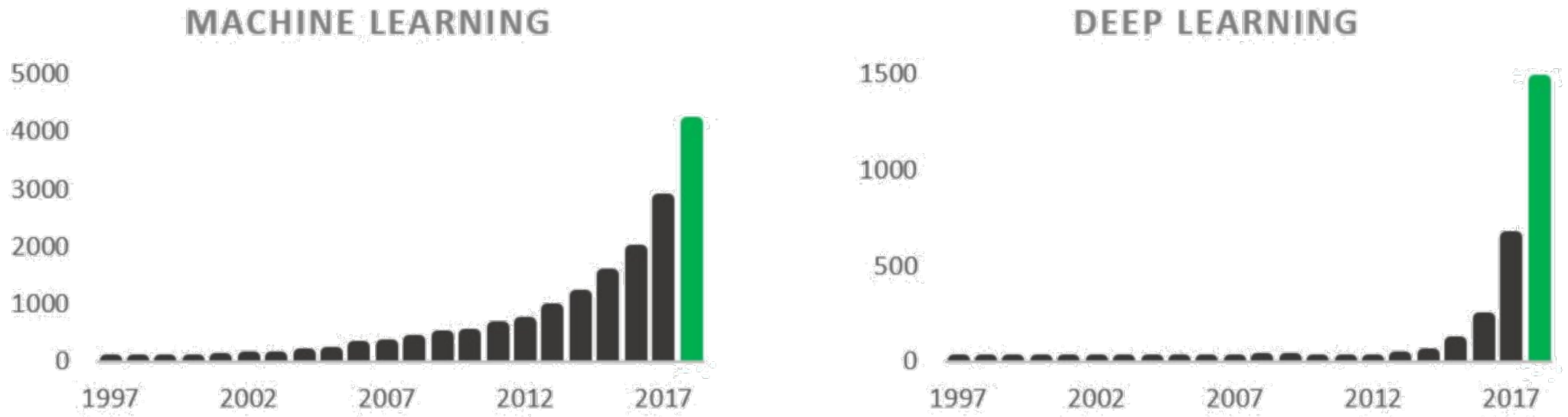








The Rise of Deep Learning

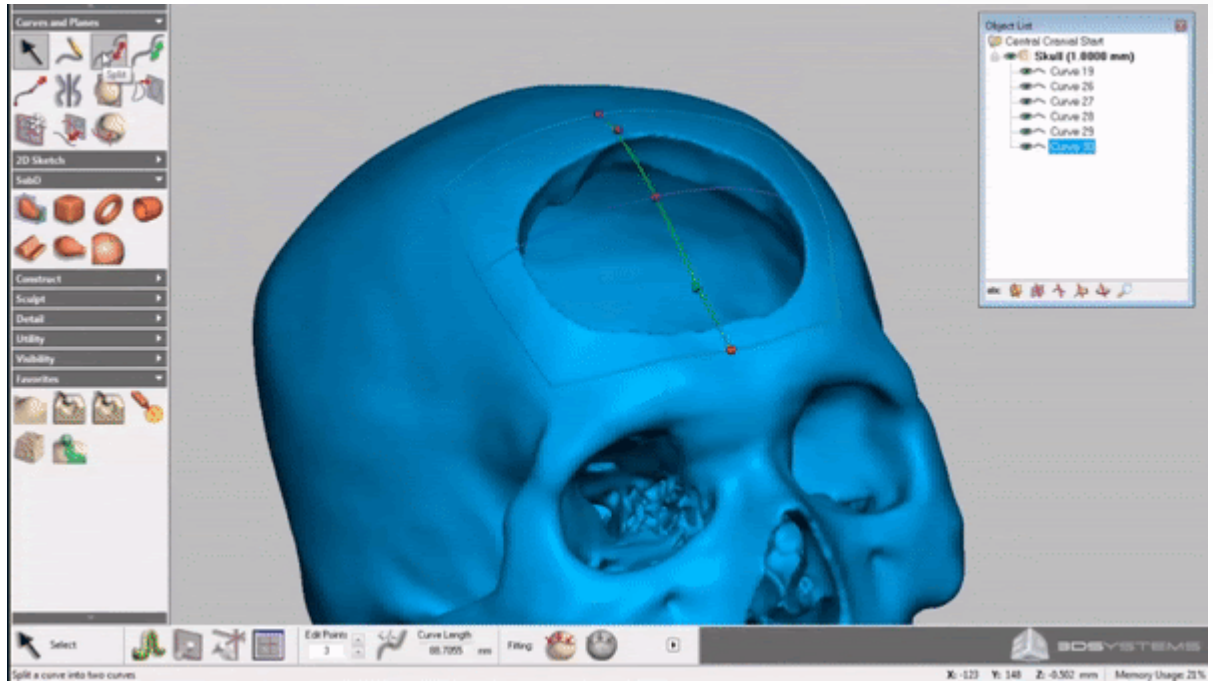


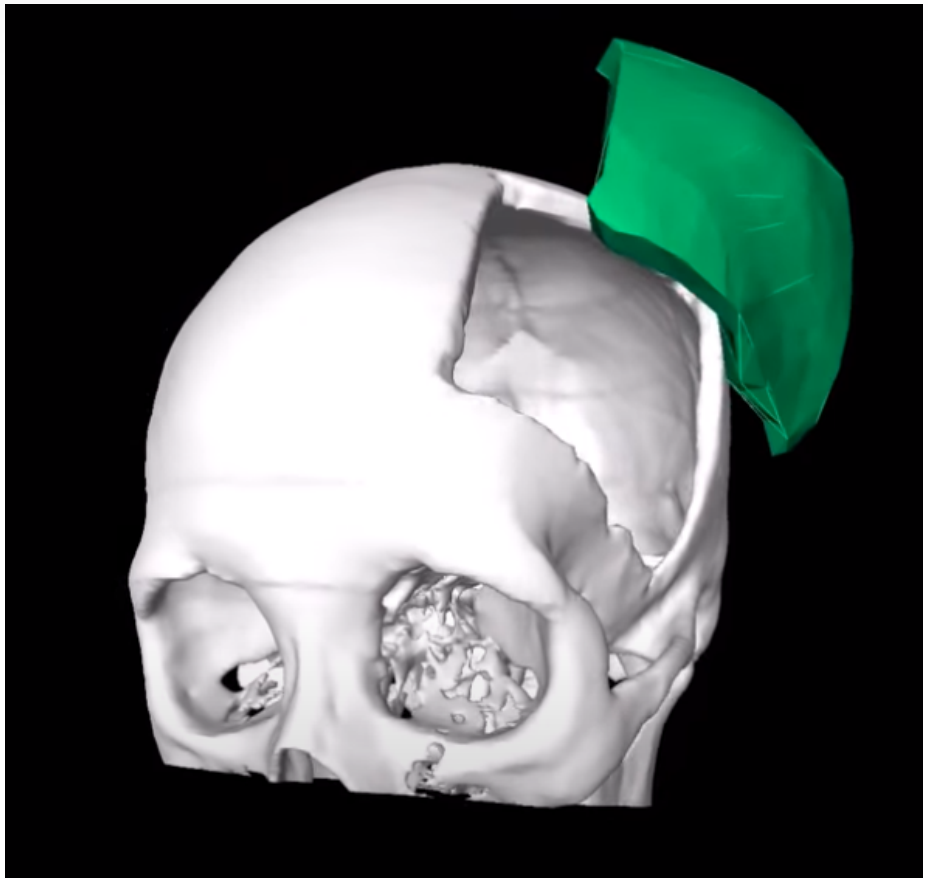
Number of publications on PubMed containing these terms in title or abstract, queried 01/11/2018

Ejemplos

Reconstrucción del cráneo para el diseño de implantes

Franco Matzkin – Actualmente becario doctoral en el sinc(i)





UNIVERSITY OF CAMBRIDGE

Reconstrucción del cráneo para el diseño de implantes

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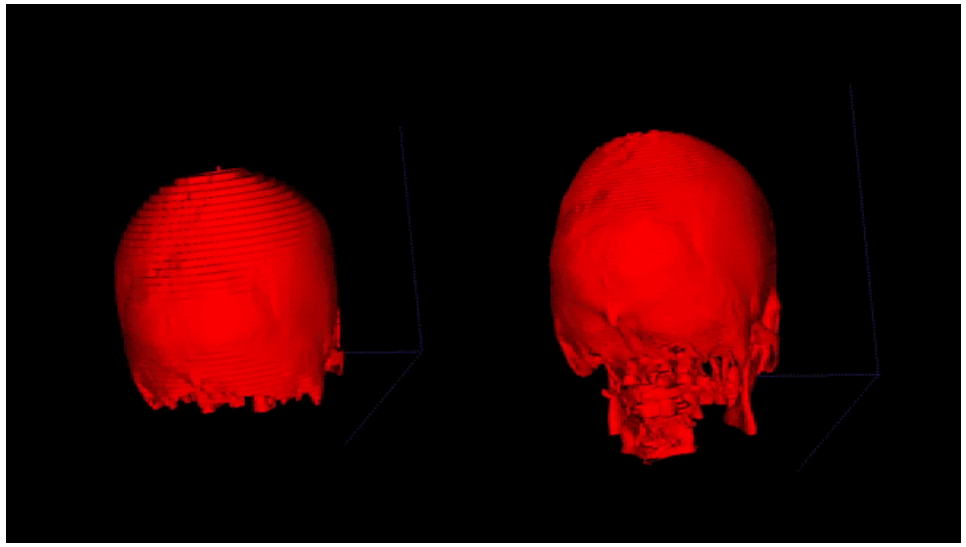
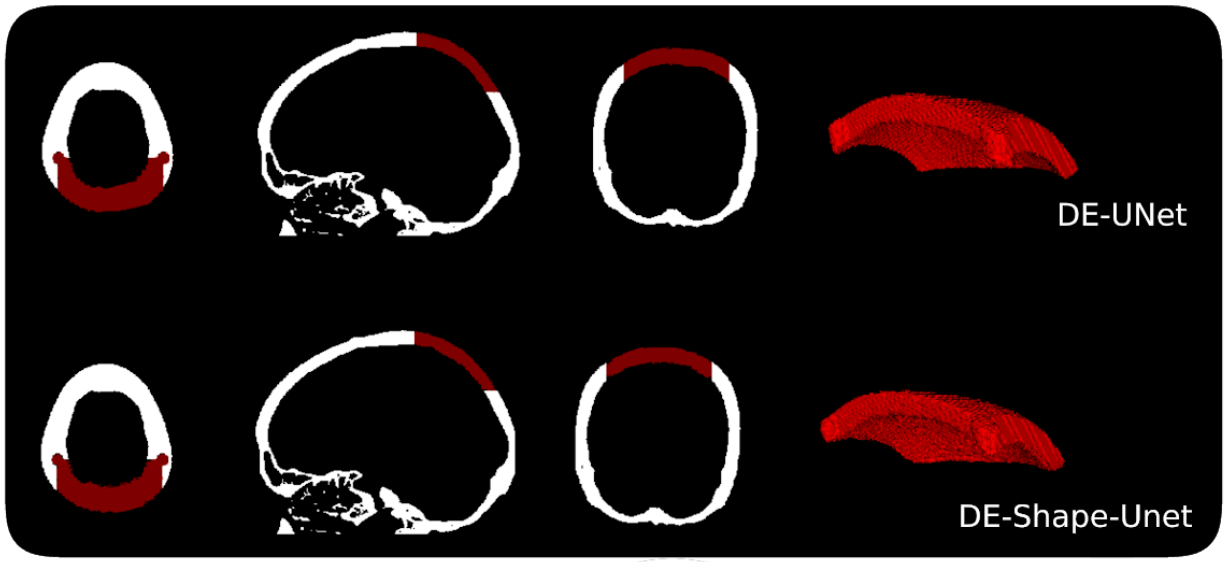
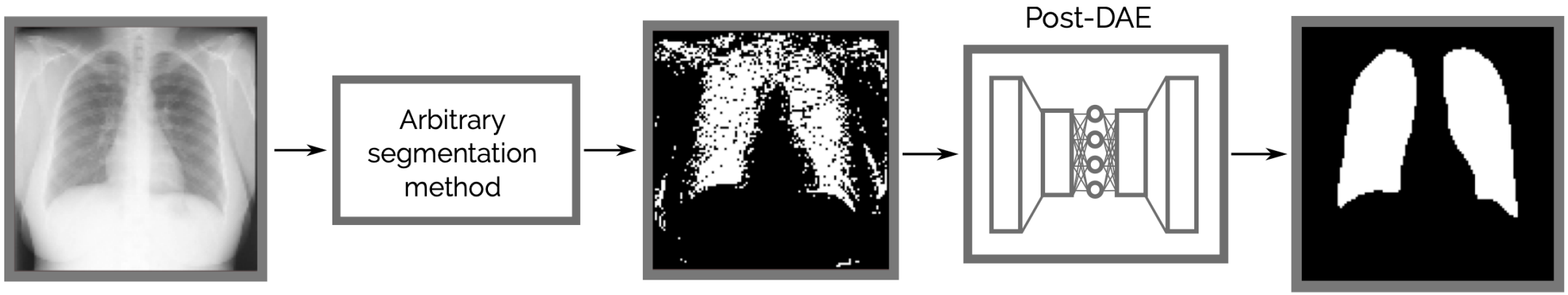




Segmentación de estructuras anatómicamente realistas

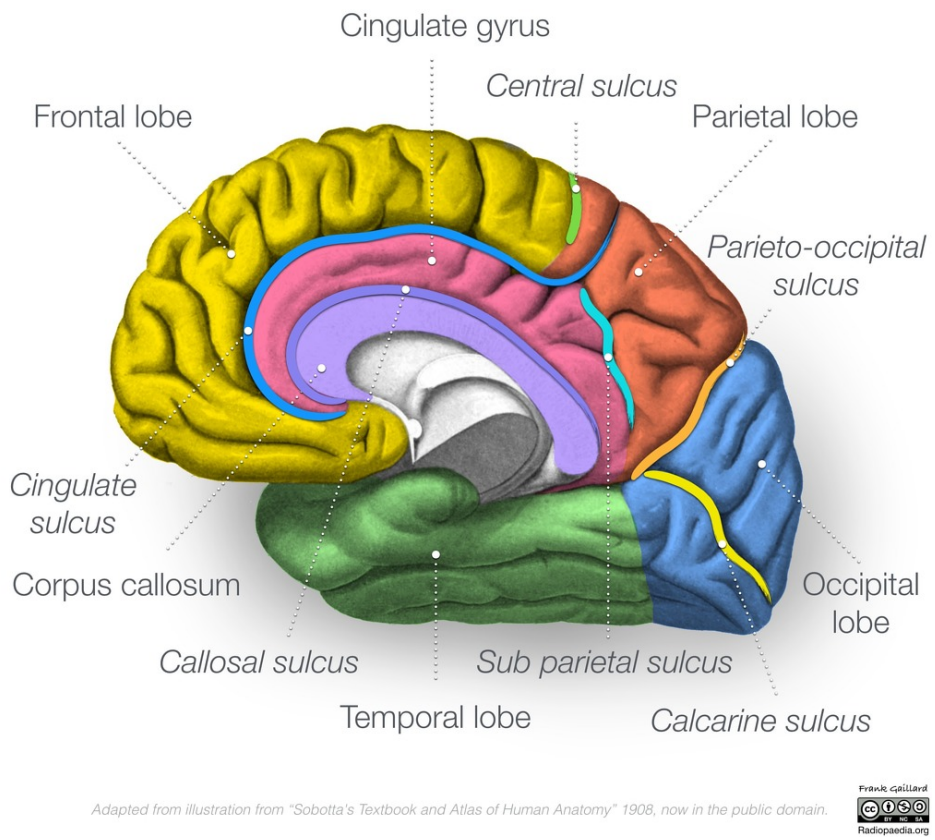
Agostina Larrazabal – Actualmente becaria doctoral en el sinc(i)

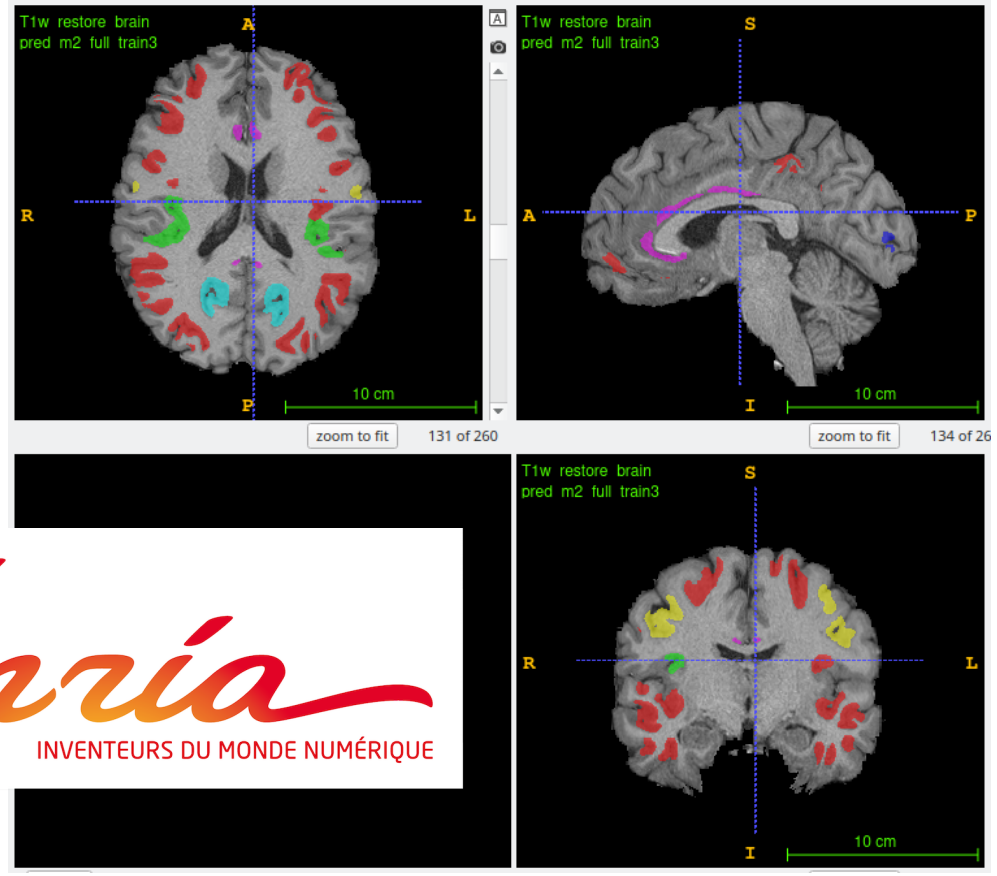


ÉTS Le génie pour l'industrie
ÉCOLE DE TECHNOLOGIE SUPÉRIEURE
Université du Québec

Segmentación y descubrimiento de surcos cerebrales

Lautaro Gianinetto – Proyecto Final Ingeniería Informática (UNL, 20xx)

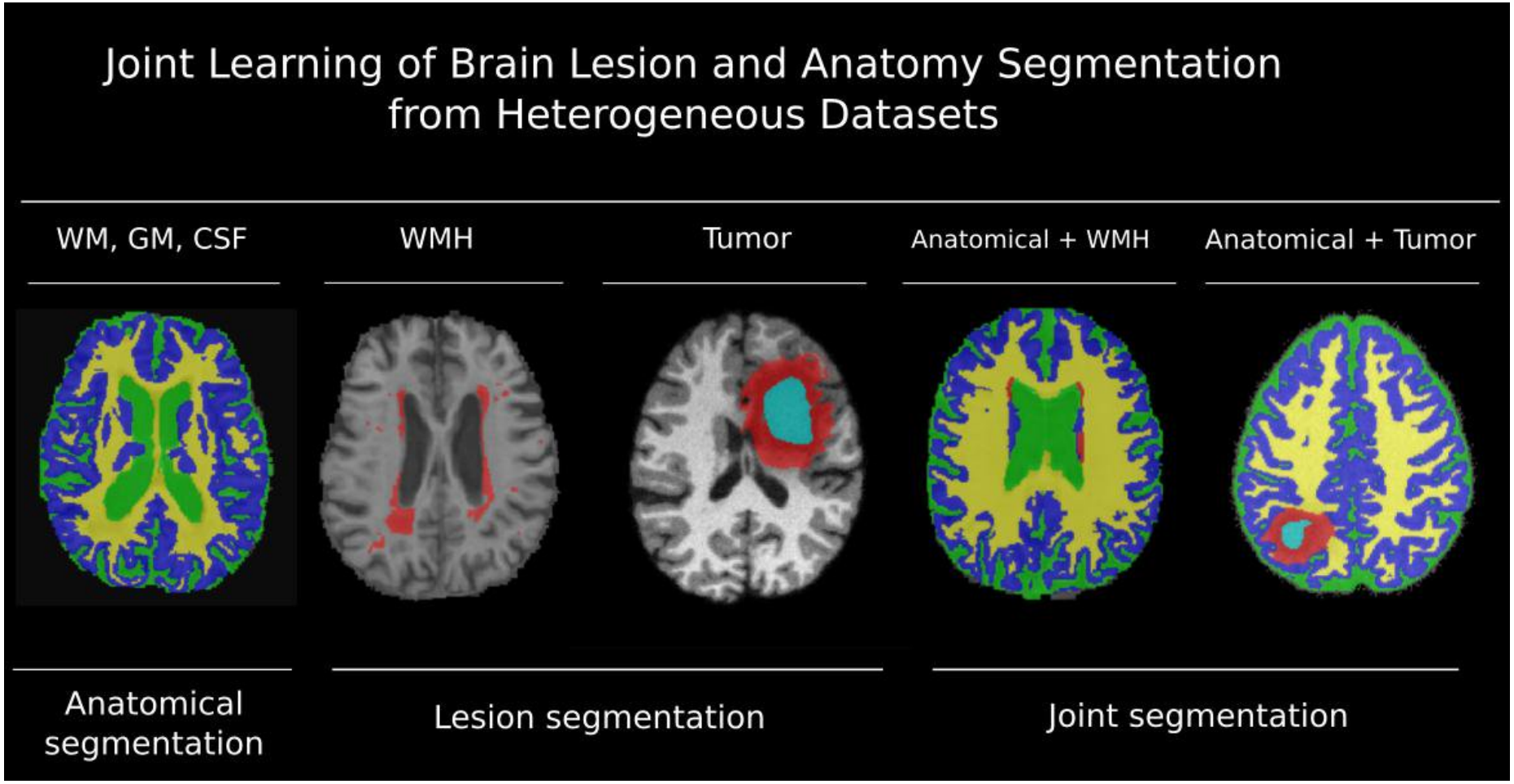




INVENTEURS DU MONDE NUMÉRIQUE

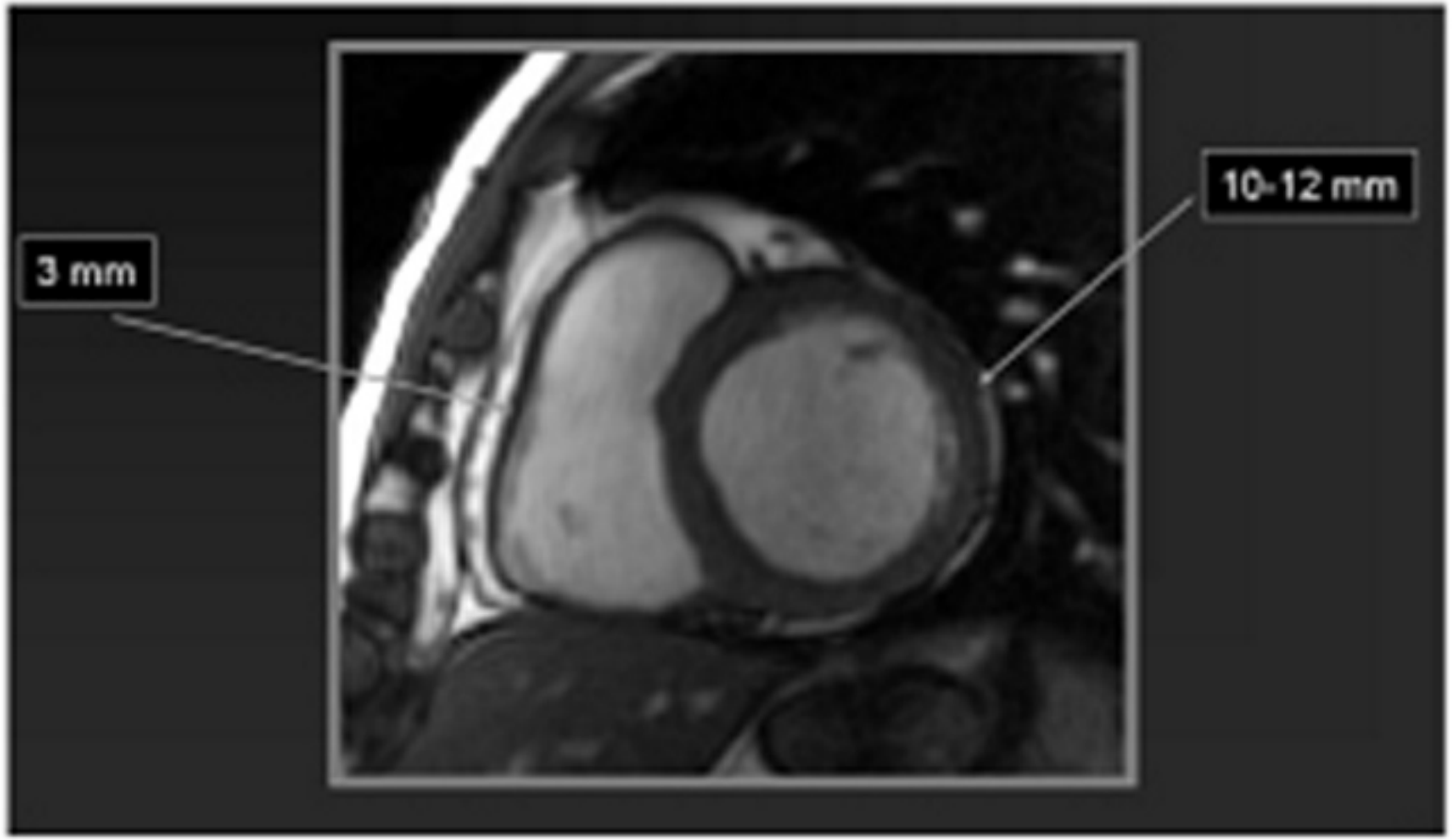
Segmentación conjunta de lesiones y estructuras anatómicas en imágenes cerebrales de MRI

Nicolás Roulet – Proyecto Final Lic. en Cs. de la Computación (UBA, 2018)



Estimación directa de índices cardíacos a partir de MRI

Alejandro Debus – Proyecto Final Ingeniería Informática (UNL, 20xx)



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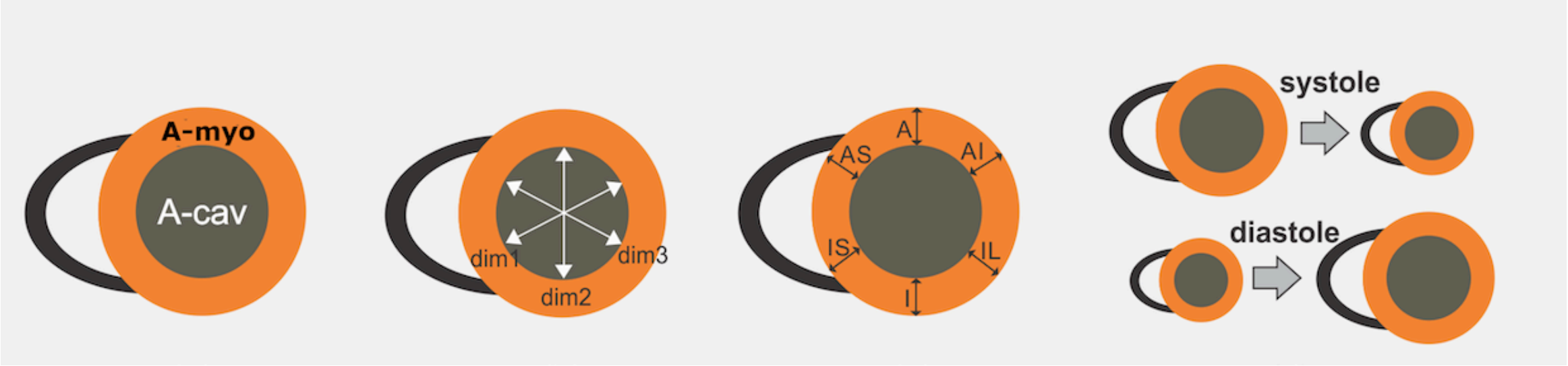
Alejandro Debus – Proyecto Final Ingeniería Informática (UNL, 20xx)



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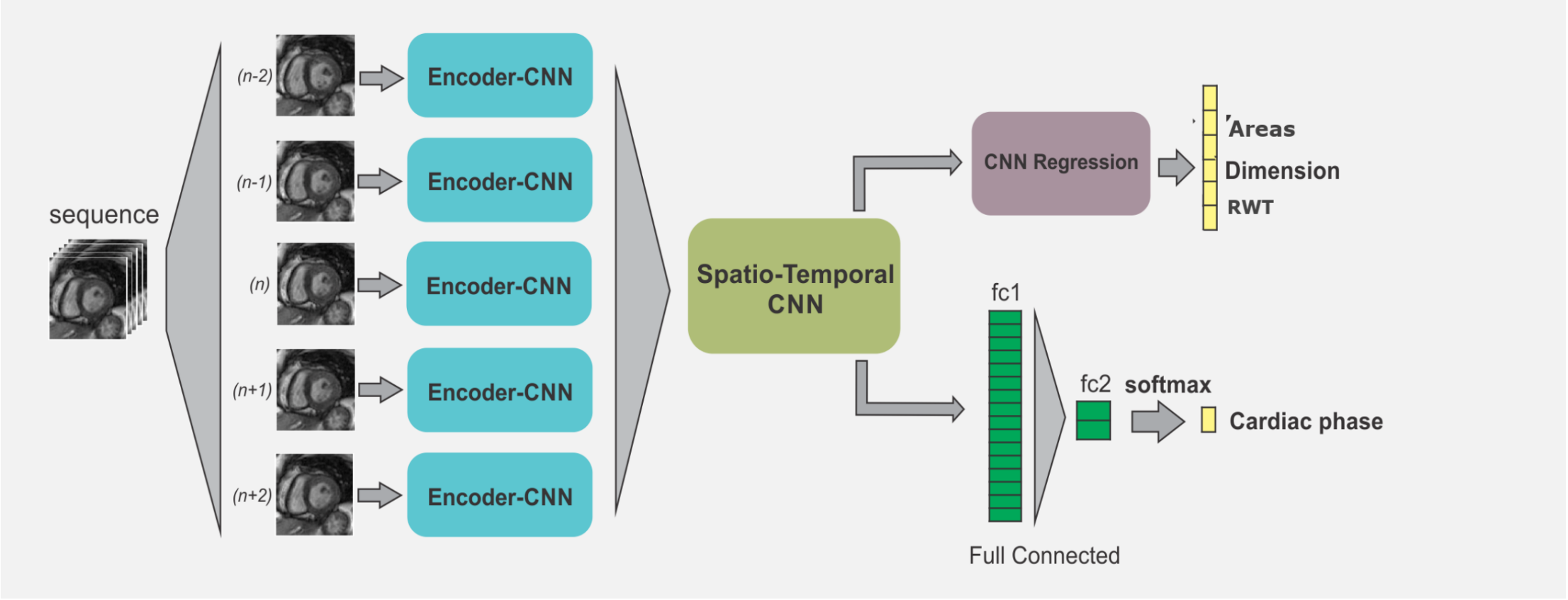
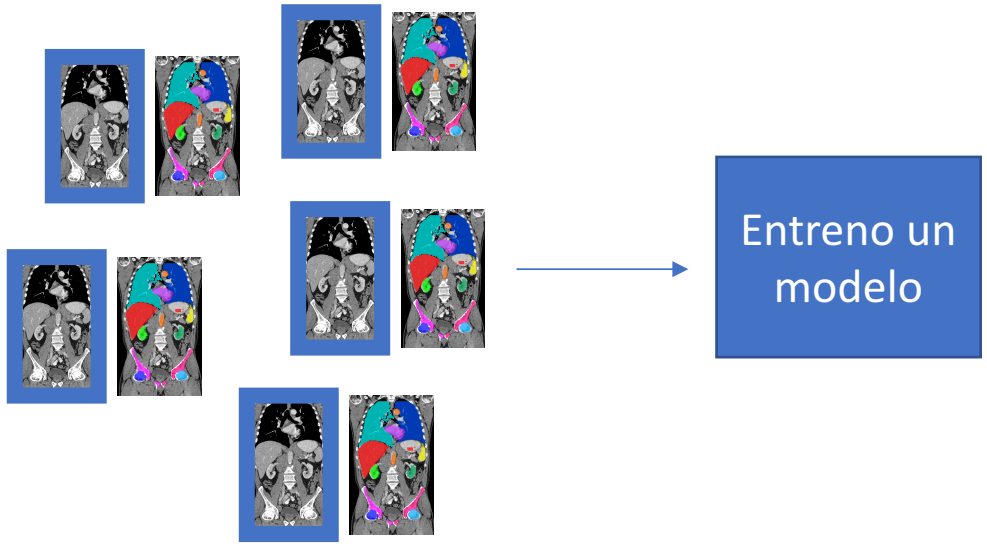


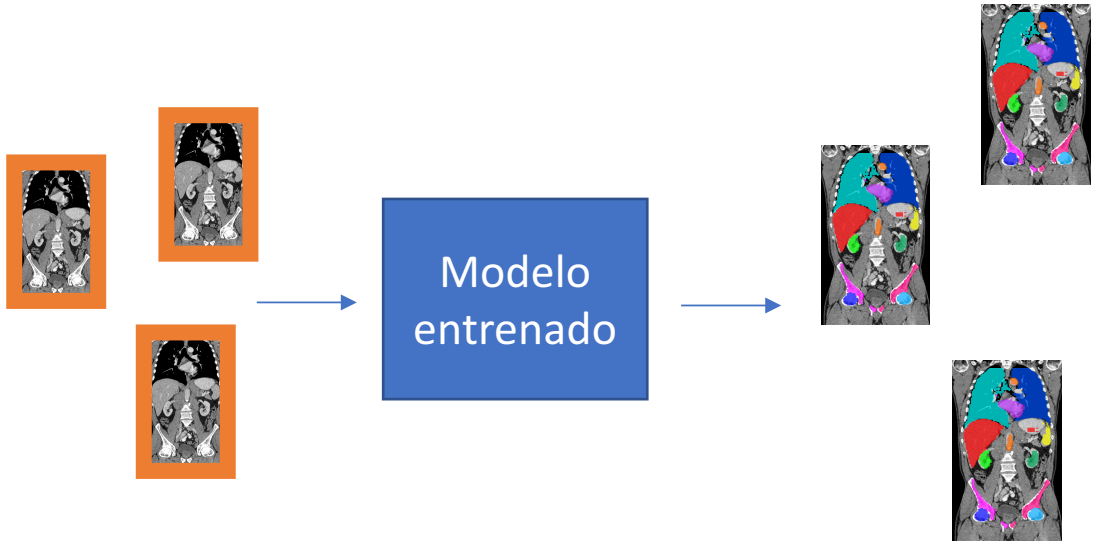
Adaptación de dominio en estudios multicéntricos

Julián Palladino – Proyecto final Lic. en Cs. de la Computación (UBA, 2020)

Lucas Mansilla – Actualmente becario doctoral en el sinc(i)



Imágenes con anotaciones obtenidas en el sitio 1



Imágenes obtenidas en el sitio 2

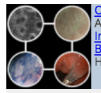
Disminución en la calidad de los resultados

Grand Challenges in Biomedical Image Analysis

Associated with: ISBI 2016
Hosted on: grand-challenge.org

CAMELYON16

The goal of this challenge is to evaluate new and existing algorithms for automated detection of cancer metastasis in digitized lymph node tissue sections. Two large datasets from both the Radboud University Medical Center and the University Medical Center Utrecht are provided.



Open for submissions
Associated with: The IEEE International Symposium on Biomedical Imaging (ISBI)
Hosted on: grand-challenge.org

ISBI-AIDA

ISBI Endoscopic Image Challenge

Open for submissions
Associated with: ISBI 2016
Article: Medical Image Analysis
Hosted on: grand-challenge.org

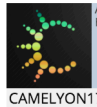
LUNA16

The LUNA16 challenge will focus on a large-scale evaluation of automatic nodule detection algorithms for chest CT.

Associated with: MICCAI 2017
Hosted on: grand-challenge.org

EndoVis

As an endoscopic vision CAI challenge at MICCAI, our aim is to provide a formal framework for evaluating the current state of the art, gather researchers in the field and provide high quality data with protocols for validating endoscopic vision algorithms.



Associated with: ISBI 2017
Hosted on: grand-challenge.org

CAMELYON17

Automated detection and classification of breast cancer metastases in whole-slide images of histological lymph node sections. This task has high clinical relevance and would normally require extensive microscopic assessment by pathologists.

Associated with: MICCAI 2017
Hosted on: grand-challenge.org

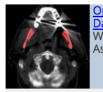
RETOUCH

The goal of the Retinal OCT Fluid Challenge is to compare automated algorithms that are able to detect and segment various types of retinal fluid lesions on a common dataset of optical coherence tomography (OCT) volumes representing different retinal diseases, acquired with devices from different manufacturers.

Open for submissions
Data download
Workshop: Oct 09, 2015
Associated with: MICCAI 2015

LV statical shape modeling: myocardial infarction

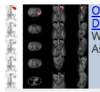
Models the statistical shape of the left ventricle (LV) to detect myocardial infarction. training comprises 100 cases with myocardial infarction 100 asymptomatic cases from the Cardiac Atlas Project.



Open for submissions
Data download
Workshop: Oct 09, 2015
Associated with: MICCAI 2015

Head and Neck Auto Segmentation

Benchmarks automatic segmentation performance. Provides 25 CT images with manually delineated data of brainstem, mandible, left and right optic nerves, optic chiasm, left and right parotid glands and submandibular glands.



Open for submissions
Data download
Workshop: Oct 09, 2015
Associated with: MICCAI 2015

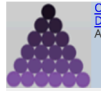
Multi-Atlas Labeling Beyond the Cranial Vault

Evaluates multi-atlas segmentation algorithms on a newly created and released standardized dataset of abdominal and pelvic anatomy on clinically acquired CT

Open for submissions
Data download

Digital Mammography DREAM Challenge

Develop algorithms to classify screening digital mammography images as low and high risk for developing cancer. Thousands of digitized mammograms and outcome data are available for training.



Open for submissions
Data download
Associated with: MICCAI 2016

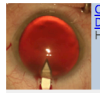
Tumor Proliferation Assessment Challenge (TUPAC)

Evaluate methods that predict the tumor proliferation score directly from whole-slide images.

Open for submissions
Data download
Associated with: MICCAI 2016

Positron Emission Tomography Segmentation (PETSEG)

Automated segmentation of PET images for the delineation of tumor volumes



Open for submissions
Data download
Hosted on: grand-challenge.org

CATARACTS

The Challenge on Automatic Tool Annotation for cataRACT Surgery aims at evaluating image-based tool detection algorithms in the context of the most common surgical procedure in the world.

Open for submissions
Data download
Hosted on: grand-challenge.org

TADPOLE

The Alzheimer's Disease Prediction Of Longitudinal Evolution (TADPOLE) challenge is brought to you by the EuroPOND consortium in collaboration with the Alzheimer's Disease Neuroimaging Initiative (ADNI).

Associated with: AAPM 2017
Hosted on: grand-challenge.org

LCTSC

This challenge will provide a platform for comparison of various automatic/semi-automatic segmentation algorithms when they are used to delineate organs at risk from CT images for thoracic radiation treatment planning.

Open for submissions
Data download
Workshop: Oct 09, 2015
Associated with: MICCAI 2015

Liver Ultrasound Tracking (CLUST 2015)

Presents the current state-of-the-art in automated tracking of anatomical landmarks in the liver and compares different methods

Open for submissions
Data download
Workshop: Oct 05, 2015
Associated with: MICCAI 2015

BraTS Multimodal Brain Image Tumor Segmentation

Publishes a large dataset of brain tumor MR scans in which the relevant tumor structures have been delineated. Each data set has T1 MRI, T1 contrast-enhanced MRI, T2 MRI, and T2 FLAIR MRI volumes

Data download
Workshop: Oct 05, 2015
Associated with: MICCAI 2015

Gland Segmentation Challenge

Gland morphology is used to assess the degree of malignancy of several adenocarcinomas, including prostate, breast, lung, and colon. This challenge evaluates gland segmentation algorithms on images of H&E stained slides, consisting of a variety of histologic grades.

Open for submissions
Data download
Associated with: MICCAI 2016

Multiple Sclerosis Segmentation (MSSEG)

MS lesions segmentation as part of the effort towards automatic segmentation of MRI scans of MS patients.

Open for submissions
Data download
Associated with: MICCAI 2016

Circuit Reconstruction from Electron Microscopy Images (CREMI)

The goal of this challenge is to evaluate algorithms for automatic reconstruction of neurons and neuronal connectivity from serial section electron microscopy data.

Open for submissions
Data download
Associated with: MICCAI 2016

Statistical Atlases and Computational Modelling of the Heart - Segmentation of Left Atrial Wall Thickness (STACOM-SLAWT)

This challenge focuses on algorithms for segmentation of left atrial wall thickness from MRI and CT.

Associated with: RSNA 2017

Pediatric Bone Age Challenge

Develop an algorithm which can most accurately determine skeletal age on a validation set of pediatric hand radiographs, using a large data set of over 12,000 images.

Associated with: MICCAI 2017

Colorectal Liver Metastases Survival Prediction

The aim of this challenge is to predict hepatic disease-free survival based on predictors derived from contrast-enhanced liver CT scans and patient clinical variables using data from Memorial Sloan Kettering Cancer Center.

Associated with: MICCAI 2017

Digital Pathology: Whole Slide Classification

This challenge focuses on classifying NSCLC patients into adeno and squamous cell cancer, and HNSCC patients into HPV +/- and K17 (mRNA) +/- molecular subtypes.

Open for submissions
Data download
Workshop: Sep 10, 2015
Associated with: BMVC 2015

CVPPP - Leaf Segmentation and Counting Challenges 2015

To advance the state of the art in leaf segmentation and to demonstrate the difficulty of segmenting all leaves in an image of plants, we organize the Leaf Segmentation and Counting Challenges (LSC and LCC).

Data download
Associated with: MICCAI 2015

ISLES 2015

Two task challenge of automatic methods for sub-acute ischemic stroke lesion segmentation and acute stroke penumbra estimation. Versatile multi-center data, two expert segmentations and extensive automatized evaluation.

Open for submissions
Data download
Associated with: ImageCLEFmed

ImageCLEFmed: Medical classification

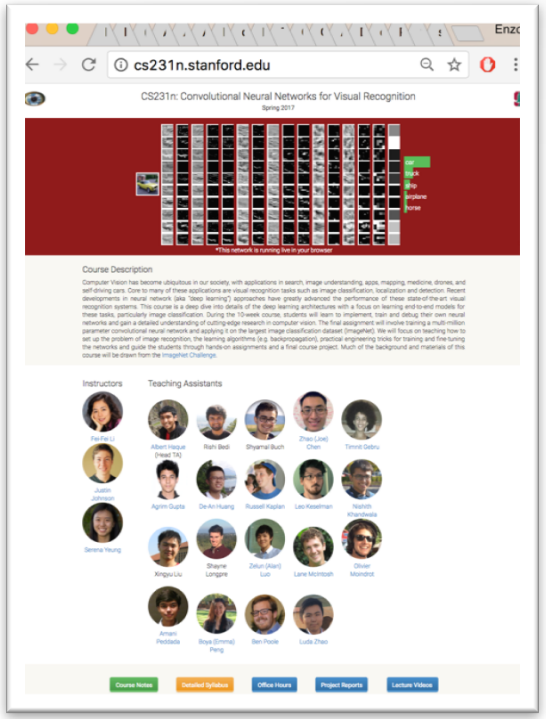
A task targeted at the identification, multi-label classifcation and separation of compound figures from biomedical literature.

https://grand-challenge.org/challenges/

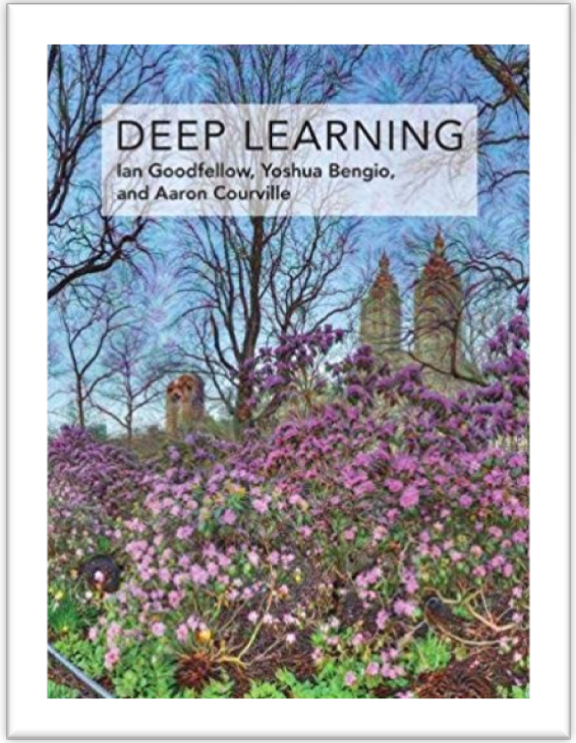
Cátedra Deep Learning 2021 – FICH - UNL

Segundo cuatrimestre 2021

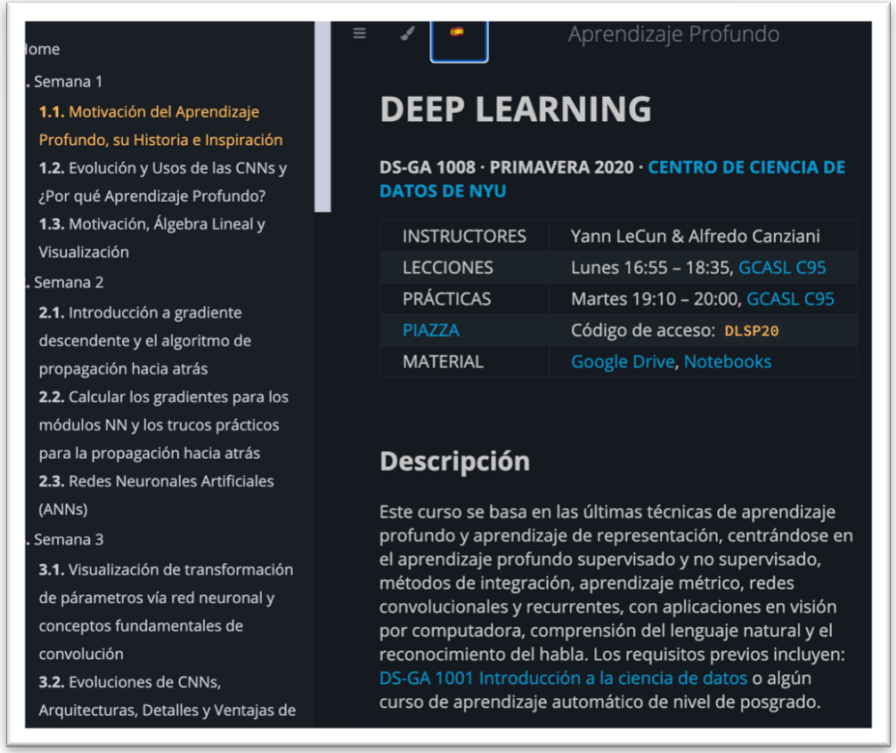
Mientras tanto…



https://cs231n.github.io/



https://www.deeplearningbook.org/



https://atcold.github.io/pytorch-Deep-Learning/

Stanford University

NYU

Aprendizaje profundo para el análisis de imágenes biomédicas

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